

Section I

Differentiation (f'(x)=?)

You do not have to simplify your answers in this exercise.

1. $8x^3 - 6x + 5$
2. $-\frac{1}{2}x^{-3/2} + \frac{1}{2}x^{-1/2}$
3. $2x - (3)(\frac{7}{3})x^{\frac{4}{3}}$
4. $1 + 0 - 2t^{-3}$
5. $-12x^{-2} - 4(-3)x^{-4} - 4x^{-5}$
6. $\frac{1}{3}x^{-2/3} - 3x^{-4}$
7. $6x - \frac{2}{3}x^{-3} + 1$
8. $\frac{1}{2}x^{-1/2} - (-\frac{1}{2})x^{-3/2}$
9. $4x^3 - 24x^2 + 4x - 1$
10. $21y^2 - 10y + 3$
11. $-27x^{-4} - 4x^{-3}$
12. $-8x^3 - 2x^{-3} - (3)(\frac{3}{4})x^{-1/4}$
13. $-8x^{-2} - (7)(\frac{1}{2})x^{-1/2} + 24x^2$
14. $9x^2 + 4x - 1$
15. $3x^2 - 3 - 2(-4)x^{-5}$
16. $1 - 4x^{-2}$
17. $(12)(4)t^3 + 9t^2 - 10t^{-3}$
18. $12x^2 - 14x + 8$
19. $(-3)(-3)u^{-4} + 8u - 2u^{-3}$
20. $f(x) = x^{3/2} + x^{-5/2}$
 $f'(x) = \frac{3}{2}x^{1/2} - \frac{5}{2}x^{-7/2}$
21. $5(x^3 + 7x^2 + 5)^4(3x^2 + 14x)$

22. $\frac{1}{2}(x^4 + x^2 + 2)^{-1/2}(4x^3 + 2x)$
23. $\frac{1}{2}(t^2 + 2t + 3)^{-1/2}(2t + 2)$
24. $2x(x^3 - 1) + x^2(3x^2)$
25. $\sqrt{1 - x^2} + x[\frac{1}{2}(1 - x^2)^{-1/2}(-2x)]$
26. $(2)(3x^2 + 2) + (2x - 4)(6x)$
27. $\frac{1}{2}(x+1)^{-1/2}(x^2+1) + (\sqrt{x+1})(2x)$
28. $4(x^3 - 3x)^3(3x^2 - 3)$
29. $[3(2x-3)^2(2)][(4x+2)^2 + (2x-3)^3[(2)(4x+2)(4)]]$
30. $[2(7x+3)(7)](3x^2 - 14x + 5) + (7x+3)^2(6x - 14)$
31. $(24y^2)(3y^2 - 5y + 10)^2 + (8y^3 - 2)[2(3y^2 - 5y + 10)(6y - 5)]$
32. $\frac{2}{3}(2x^3 - 3)^{-1/3}(6x^2)$
33. $2(3x+4) + (2x-3)(3)$
34. $6x^2(x^4 + x) + (2x^3 - 1)(4x^3 + 1)$
35. $2(2x+1)(2)(x^2+2)^3 + (2x+1)^2(3)(x^2+2)^2(2x)$
36. $\frac{1}{2}(3x^2+1)^{-1/2}(6x)$
37. $\frac{1}{2}(3x^2-2x+1)^{-1/2}(6x-2)$
38. $(-x^{-2})(x^2-5) + (\frac{1}{x}+3)(2x)$
39. $3(2t-5)^2(2)$
40. $-3(6x-5)^{-4}(6)$
41. $\frac{(2x+1)(x^2+1) - (x^2+x+1)(2x)}{(x^2+1)^2}$
42. $\frac{(2)(3x+2) - (2x+3)(3)}{(3x+2)^2}$
43. $\frac{(2)(3u-5) - (2u+1)(3)}{(3u-5)^2}$

$$44. \frac{(2x+5)(x^2) - (x^2+5x-1)(2x)}{(x^2)^2}$$

$$45. f(x) = \frac{(x+1)(x+2)}{(x-1)(x-2)} \\ = \frac{x^2+3x+2}{x^2-3x+2}$$

$$f'(x) = \frac{(2x+3)(x^2-3x+2) - (x^2+3x+2)(2x-3)}{(x^2-3x+2)^2}$$

$$46. \frac{1}{2} \left(\frac{x^2+1}{x^2+4} \right)^{-1/2} \left(\frac{(2x)(x^2+4) - (x^2+1)(2x)}{(x^2+4)^2} \right)$$

$$47. \frac{(1+3t^2)(\sqrt{t}) - (t+t^3)\left(\frac{1}{2\sqrt{t}}\right)}{(\sqrt{t})^2}$$

$$48. (2) \left(\frac{x+1}{x-1} \right) \left(\frac{(1)(x-1) - (x+1)(1)}{(x-1)^2} \right)$$

$$49. f(x) = \frac{10}{\sqrt{x}+4} = 10(\sqrt{x}+4)^{-1}$$

$$f'(x) = -10(\sqrt{x}+4)^{-2} \left(\frac{1}{2\sqrt{x}} \right)$$

$$50. f(x) = \frac{8}{4+x^2} = (8)(4+x^2)^{-1}$$

$$f'(x) = -8(4+x^2)^{-2}(2x)$$

$$51. \frac{4(y^2+1) - (4y)(2y)}{(y^2+1)^2}$$

$$52. f(x) = \frac{3}{\sqrt{2x+1}} = 3(2x+1)^{-1/2}$$

$$f'(x) = -\frac{3}{2}(2x+1)^{-3/2}(2)$$

$$53. \frac{(6x-2)(4x^2-5) - (3x^2-2x+3)(8x)}{(4x^2-5)^2}$$

$$54. f(x) = \frac{(2x-4)(3x+5)}{2x^2+7}$$

$$= \frac{6x^2-2x-20}{2x^2+7}$$

$$f'(x) = \frac{(12x-2)(2x^2+7) - (6x^2-2x-20)(4x)}{(2x^2+7)^2}$$

$$55. \frac{(2)(x^2+2x) - (2x-3)(2x+2)}{(x^2+2x)^2}$$

$$56. \frac{(-2+6t)(t^2+2) - (4-2t+3t^2)(2t)}{(t^2+2)^2}$$

$$57. 6x^2 + \frac{(-1)(x^3) - (2-x)(3x^2)}{(x^3)^2}$$

$$58. \frac{(1)(x+1) - (x-1)(1)}{(x+1)^2}$$

$$59. \frac{(2x)(x^2+1) - (x^2)(2x)}{(x^2+1)^2}$$

$$60. f(t) = (t^4-2t+1)^{-1}$$

$$f'(t) = (-1)(t^4-2t+1)^{-2}(4t^3-2)$$

$$61. [\cos(x^2+2)](2x)$$

$$62. (\cos(3x))(3)$$

$$63. (-\sin(3y))(3)$$

$$64. [-\sin(x^4+7)](4x^3)$$

$$65. [\sec^2(\sin x)](\cos x)$$

$$66. [\sec^2(t^2+5)](2t)$$

$$67. (\cos(3x+2))(3)$$

$$68. 2\cos x - \sec^2 x$$

$$69. 1 + \sin x$$

$$70. \frac{(\cos x)(x) - (\sin x)(1)}{x^2}$$

$$71. [\sec^2(2x-x^3)](2-3x^2)$$

$$72. [\cos(3\theta^2-2\theta+1)](6\theta-2)$$

$$73. 3\cos x + 2\sin x$$

$$74. [\sin(\pi x-1)](\pi)$$

$$75. 2x \sin x + x^2 \cos x$$

$$76. [-\sin(\tan x)](\sec^2 x)$$

$$77. [3\cos(x^2)](2x) + 2\cos x - 0$$

$$78. [-\sin(3x)](3) - [\sec^2(3x)](3)$$

$$79. [\cos(2x+3)](2)$$

$$80. [\cos(2\pi t)](2\pi)$$

$$81. [\cos(\frac{1}{\sqrt{x}})](-\frac{1}{2}x^{-3/2})$$

$$82. [\cos(\frac{1}{5x})](-x^{-2})$$

$$83. [-\sin(-x)](-1)$$

$$84. [\sec^2 \pi (\frac{1}{2} - x)](-\pi)$$

$$85. [\sec^2(x^2+1)](2x)$$

$$86. 5 \sec^2(5t)$$

$$87. \cos x - x \sin x$$

$$88. \frac{\cos x(1+x) - \sin x(1)}{(1+x)^2}$$

$$89. [8 \sin(3x)](\cos(3x))(3)$$

$$90. \frac{(-\sin x)(\sin x) - (\cos x)(\cos x)}{(\sin x)^2} \\ = -\frac{(\sin^2 x + \cos^2 x)}{\sin^2 x} = \frac{-1}{\sin^2 x}$$

$$91. 2x \sin x + (x^2+3) \cos x$$

$$92. \frac{1}{2} (\sin x)^{-1/2} \cos x$$

$$93. f(x) = [\cos(x^3)]^2$$

$$f'(x) = 2[\cos(x^3)]^2 (-\sin(x^3))(3x^2)$$

$$94. (1) \cos(5x) - (5x) \sin(5x)$$

$$95. 2 \cos(2y) \cos(3y) - 3 \sin(2y) \sin(3y)$$

$$96. (2)(3) [\sin(3x)] \cos(3x) + \\ (2)(5) [\cos(5x)] (-\sin(5x))$$

$$97. 2x \tan(\frac{1}{x}) + x^2 \sec^2(\frac{1}{x})(-\frac{1}{x^2})$$

$$98. (-1)(\sin x - x \cos x)^{-2} x \\ [\cos x - \cos x + x \sin x]$$

$$99. \frac{(\cos x - \sin x) \tan x - (\sin x + \cos x) \sec^2 x}{(\tan x)^2}$$

$$100. f(x) = \sin^2(x + \frac{1}{x}) + \cos^2(x + \frac{1}{x}) = 1 \\ f'(x) = 0$$

$$101. \frac{(2x)(\tan x) - x^2 \sec^2 x}{(\tan x)^2}$$

$$102. \frac{1}{2} (\cos(2x))^{-1/2} (-2 \sin(2x))$$

$$103. 2 \sin(x-3) \cos(x-3)$$

$$104. 3 \cos^2(t^2-t) (-\sin(t^2-t))(2t-1)$$

$$105. \cos x \cos^2 x + \sin x (2 \cos x) (-\sin x)$$

$$106. [\frac{1}{3} \sin^{-2/3}(2x)] (\cos 2x) (2)$$

$$107. [28 \sin^6(2-4x)] \cos(2-4x) (-4)$$

$$108. 2 \cos x + 6 \sin x \cos x$$

$$109. -9 \cos^2 u \sin u - 8 \cos u \sin u$$

$$110. -10 \cos(x+2) \sin(x+2) - 3 \sin(6+2)$$

$$111. 10x - 6 \tan x \sec^2 x + \cos x$$

$$112. \frac{\sec^2 x (1 - \tan x) - (\tan x) (-\sec^2 x)}{(1 - \tan x)^2}$$

$$113. \frac{3}{2} (\sin(x+1))^{1/2} \cos(x+1)$$

$$114. f(t) = \sin^2 t + \cos^2 t = 1 \\ f'(t) = 0$$

$$115. (\cos \sqrt{x}) (\frac{1}{2\sqrt{x}}) + \frac{1}{2} (\sin x)^{-1/2} \cos x$$

$$116. (\cos x) (\sin x + \cos x) + \\ (\sin x) (\cos x - \sin x)$$

$$117. f(x) = \sin 2x$$

$$f'(x) = 2 \cos 2x$$

$$118. -(\sin u)^{-2} (\cos u)$$

$$119. -2 \cos x \sin^2 x + \cos^3 x$$

$$120. f(x) = \sqrt{\cos^2 x} \\ f'(x) = \frac{1}{2} (\cos^2 x)^{-1/2} (2)(\cos x)(-\sin x)$$

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121. $1 + 2 \tan x \sec^2 x$

122. $\frac{(1) \cos x + x \sin x}{(\cos x)^2}$

123. $\sec^2 x \cos x - \tan x \sin x$

124. $\cos^2 t - \sin^2 t$

125. $(e^{x^2})(2x)$

126. $5e^{5x}$

127. $(2x+3)e^x + (x^2+3x)e^x$

128. $e^x + xe^x + e^{-x}$

129. $f(x) = e^{x^2+x+1}$

$$f'(x) = (e^{x^2+x+1})(2x+1)$$

130. $f(x) = e^{x^2-x+1}$

$$f'(x) = (e^{x^2-x+1})(2x-1)$$

131. $-e^{-u}$

132. $\frac{2}{3}e^{\frac{2x}{3}}$

133. $(e^{5x})(\frac{1}{2\sqrt{x}})$

134. $3e^{3x} + 4e^{2x} - 3e^x + 0$

135. $(e^{x^2-2})(2x)$

136. $\frac{2e^{2x}(2-e^{2x}) - (1+e^{2x})(-2e^{2x})}{(2-e^{2x})^2}$

137. $3e^{3x-1} + 4e^{-x}$

138. $[-\sin(e^t)](e^t)$

139. $6e^{2x} - 4e^x + 0$

140. $(e^{3 \cos(2x)})(3)(-\sin(2x))(2)$

141. $-2e^{-2x} - 12e^{-3x} + 0$

142. $2e^{2x+1}$

143. e^{2x}

144. $(e^{\sin x})(\cos x)$

145. $2e^{2x}$

146. $2e^t + 2te^t$

147. $f(x) = (1-e^{-x})^{-1}$

$$f'(x) = -(1-e^{-x})^{-2}(e^{-x})$$

148. $\frac{-xe^{-x} - e^x(1)}{x^2}$

149. $2xe^{-x} - x^2e^{-x}$

150. $(e^{-\frac{1}{x^2}})(2)(\frac{1}{x^3})$

151. $(e^{\sqrt{x^2+1}})(\frac{2x}{2\sqrt{x^2+1}})$

152. $\frac{1}{2}(e^x + e^{-x})$

153. $2e^{2x} \cos(3x) - 3e^{2x} \sin(3x)$

154. $(e^{\cos(4x)})(-4 \sin(4x))$

155. $(2x)(2^x) + (x^2)(2^x \ln 2)$

156. $(3^{5x} \ln 3)(5)$

157. $4x^3 + 4^x \ln 4$

158. $-9^{-x} \ln 9$

159. $(\sec^2(5x))(5^x \ln 5)$

160. $(3^{4x+1} \ln 3)(4) + (2^{4x+2} \ln 2)(4)$

161. $(3^{y^2+1} \ln 3)(2y)$

162. $2^x \ln 2$

163. $-2^{-x} \ln 2$

164. $(\frac{1}{2})^6 \ln(\frac{1}{2})$

165. $e^x \ln x + (e^x)(\frac{1}{x})$

166. $\frac{\cos x}{\sin x} = \cot x$

167. $-(\ln x)^2(\frac{1}{x})$

168. $f(x) = \ln(3x-x)$

169. $\frac{1}{x-1} - \frac{2x}{x^2+1}$

170. $f'(x) = \frac{3}{3x} - 1$

171. $\frac{d}{dx}(\sin 2x) = 2 \cos 2x$

172. $\frac{1}{2}[\frac{1}{x} - \frac{2x}{x^2+1}]$

173. $\frac{2u}{u^2}$

174. $\frac{1}{2x}$

175. $\ln 10$

176. $\frac{3}{3x} + \frac{4}{x} = \frac{5}{x}$

177. $2x \ln(2x) + (x^2)(\frac{2}{2x})$

178. $\frac{d}{dx}(-\ln x) = -\frac{1}{x}$

179. $\ln x + 1$

180. $-\frac{1}{x}$

181. $3(\ln y)^2(\frac{1}{y})$

182. $\ln \sqrt{x} + (x)(\frac{1}{2})(\frac{1}{x})$

183. $\frac{7}{7x}$

184. $\frac{1}{2}(\ln x)^{-\frac{1}{2}}(\frac{1}{x})$